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Photovoltaic Performance Investigation of $Cs_3Bi_2I_9$-Based Perovskite Solar Cells with Various Charge Transport Channels Using DFT and SCAPS-1D Frameworks

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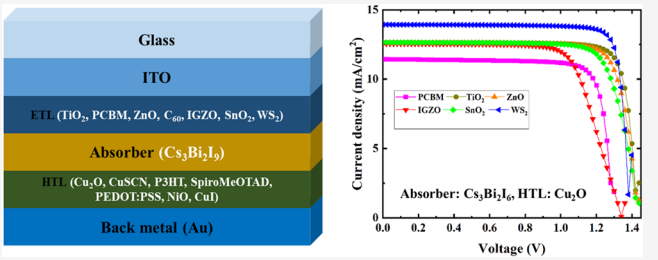
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ABSTRACT: $Cs_3Bi_2I_9$ as a solar absorber material is a strong contender for lead-free perovskite solar cells (PSCs). The presence of bismuth (Bi) in $Cs_3Bi_2I_9$ leads to the origin of interesting optoelectronic properties along with a suitable optical band gap and high absorption coefficient. However, further analysis of the crystal structure, optical, and electronic properties of this material is required for efficient photovoltaic (PV) applications. The potential of $Cs_3Bi_2I_9$ perovskite as an absorber layer for solar cells (SCs) was first analyzed by performing density functional theory (DFT) calculations to observe its structural, optical, and electronic properties. Band structure reveals an indirect band gap (2.42 eV), and density of states (DOS) data show good conductivity primarily contributed by the 5p and 6s orbital electrons of I and Bi atoms. Strong electronic charge buildup is seen in the electronic charge density map surrounding the I atom, as well as the covalent bonds between the I and Bi atoms. The frequency-dependent dielectric function and absorption calculations reveal that $Cs_3Bi_2I_9$ might be a potential material in optoelectronic and photovoltaic systems. We also performed numerical simulations using the one-dimensional solar cell capacitance simulator (SCAPS-1D) for 49 different PSC configurations with $Cs_3Bi_2I_9$ absorber, electron transport layers (ETLs) comprising WS_2, indium−gallium−zinc oxide (IGZO), SnO_2, ZnO, C_{60}, TiO_2, and phenyl-C61-butyric acid methyl ester (PCBM), and hole transport layers (HTLs) like Cu_2O, CuSCN, NiO, poly(3-hexylthiophene) (P3HT), poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS), Spiro-MeOTAD, and CuI. Simulation results reveal that the Cu_2O HTL exhibited the best power conversion efficiency (PCE) for all of the ETLs. Of the 49 configurations, the six best configurations with the Cu_2O HTL and different ETLs were analyzed to study the effect of absorber and ETL thickness, series and shunt resistances, operating temperature, capacitance, Mott−Schottky, generation, and recombination rate on the PV performance. Current−voltage (J−V) characteristics and quantum efficiency (QE) were computed for all of these configurations to understand the impact of the absorber, ETL, and HTL on the PV parameters. This comprehensive simulation study will assist researchers in the fabrication of cheap and efficient PSCs without lead and open new horizons in the field of solar technology.



1. INTRODUCTION

In recent years, organic metal halide perovskites with the structural formula of AMX_3 (A = organic group; M = metal ion; and X = halide ions) have emerged as viable contenders for perovskite solar cell (PSC) technologies[1−4] due to their exceptional optoelectronic properties such as tunable band gap, high absorption coefficient, extended diffusion lengths, efficient charge transport, and long carrier lifetimes.[5−8] Although the lead-based perovskites show similar optical and electronic properties and have reached efficiencies exceeding 25%,[9] which is comparable to that of crystalline silicon, their commercial application is still questionable due to the two major issues of toxicity and stability.[10] These two factors drove the researchers to look for alternative materials for future PSCs considering the feasibility of commercialization.[11−15] Initially, tin (Sn) and germanium (Ge) were considered effective replacements for Pb-based perovskites due to the similarity in their valence shell electronic configuration (all three being group 14 elements).[16] However, the PCEs of Sn- and Ge-based PSCs

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are limited by their chemical and atmospheric instabilities in the desired oxidation state.[17]

As a result of the above-mentioned problems, a new class of Bi-based perovskites with the chemical formula $A_3Bi_2X_9$ (where A refers to a monovalent organic or inorganic cation and B is a halogen atom) have gained popularity due to the closeness of Bi and Pb in the periodic table for possessing ns^2 electrons.[18] Additionally, Bi-based perovskites show excellent stability under humid atmospheric conditions and can be manufactured easily using cost-effective solution processing techniques at low temperatures.[19] However, the efficiency of Bi-based halide PSCs is much less as compared to their Pb counterparts due to their higher electron−hole pair-binding energy and low absorption coefficient, thereby resulting in poor photocurrent density.[18,20] For this reason, there are limited reports on the experimental synthesis of this perovskite family, which demands rigorous analysis of the SC architectures involving the $Cs_3Bi_2I_9$ absorber.

Park et al. investigated the photovoltaic performance of $Cs_3Bi_2I_9$ and $MA_3Bi_2I_9$ and reported an efficiency of ∼1% for $Cs_3Bi_2I_9$ PSCs.[21] The zero-dimensional molecular structure of $Cs_3Bi_2I_9$ resulting in inferior charge transport is considered to be one of the major reasons for the poor PV performance, and it is proposed that an enhancement in PCE can be achieved by utilizing the 3D crystal structure of $Cs_3Bi_2I_9$.[18] A higher efficiency of ∼3.6% is reported in bulk heterojunction Bi-based PSCs consisting of phase-separated $Cs_3Bi_2I_9$ and $Ag_3Bi_2I_9$ as the photoactive layer. Type-II band alignment and optimized crystal structure and crystal orientation resulting in efficient exciton separation and charge transport were the main factors contributing to the enhanced performance.[22] $Cs_3Bi_2I_9$ nanosheet-based PSCs with CuI as the HTL exhibited a remarkable PCE of 3.20% along with excellent prolonged stability in ambient air, maintaining 57% of the initial PCE after 38 days of exposure.[8] Although the PCE of 2.81% reported in ref 20 is slightly lower than the above-mentioned reports, it was found that the addition of potassium iodide (KI) can be beneficial in improving device stability against humidity. All of these inventions brought a breakthrough in the research on $Cs_3Bi_2I_9$ absorber, generating a lot of interest in the theoretical analysis of different configurations of PSCs with the help of the one-dimensional solar cell capacitance simulator (SCAPS-1D) over the past two years.[23−26]

For use in photovoltaic (PV) devices, computational and experimental methodologies were employed to investigate the structural properties, electronic structures, and optical characteristics of $A_3Bi_2I_9$ (A = Cs, K, Rb) perovskites and their different phases. Theoretical investigations using density functional theory (DFT) have satisfactorily described the special characteristics of organic−inorganic hybrid perovskites.[27,28] First principle calculations using the VASP code show that $Cs_3Bi_2I_9$ exhibits an indirect band gap of 2.32 eV and an indirect band gap of 2.43 eV as computed using the PBE functional[29] in this study. Using first principle calculations, Ghosh et al. predicted that the poor PV performance of $Cs_3Bi_2I_9$ is mainly due to its indirect band gap nature, high carrier effective masses, and deep-level defect states, which act as effective recombination centers.[30] It is well-known that direct band gap materials are generally preferred for optical applications because of their efficient light emission and absorption properties. Besides, indirect band gap materials may be cheaper to fabricate, offer higher flexibility, have unique electronic properties, and be achieved easily for PV applications. Apart from this, most of the computational studies on $Cs_3Bi_2I_9$ are mainly focused on its phase transitions and crystal structures.[31,32] However, except for a few estimated parameters for $Cs_3Bi_2I_9$,[33] the examination of the characteristics of inorganic bismuth halides has been restricted to some studies describing optical and electronic properties.[34,35] Therefore, it is really important to evaluate the functionality and role of various atoms in $Cs_3Bi_2I_9$.

The efficiency and stability of PSCs can be controlled to a large extent by modulating the quality of different functional layers in the n-i-p or p-i-n device stacks. To boost the PCE of PSCs, combinations of different electron transport layers (ETLs) and hole transport layers (HTLs) are investigated. For enhancing the device performance and reliability, the heterostructure interface between the ETL and perovskite plays a crucial role.[36] ETLs like aluminum oxide (Al_2O_3),[37] zinc oxide (ZnO),[38−40] zirconium oxide (ZrO_2),[41] tin oxide (SnO_2), and titanium dioxide (TiO_2)[42] are some of the metal oxides that have been used in the of PSCs for improvement in PV performance. In addition to these, indium−gallium−zinc oxide (IGZO),[43] tungsten disulfide (WS_2),[44,45] ceric dioxide (CeO_2),[46] phenyl-C61-butyric acid methyl ester (PCBM),[47] and C_{60}[48] are also effective as ETLs.

In a similar way, HTLs also promote the effective extraction of photogenerated holes from the perovskite and the transport of these holes to the cathode, thereby affecting the device's performance and durability directly. It is possible to categorize HTLs into inorganic, small molecule, and polymeric subsets based on their molecular structures. Over the years, HTLs like Cu_2O, CuSCN, NiO, P3HT, PEDOT:PSS, Spiro-MeOTAD, and CuI gained popularity due to their application and performance in various studies.[42,49−53] Out of all of these HTLs, Cu_2O is a nontoxic p-type semiconductor with a direct gap of ∼2.1 eV.[42] Cu_2O thin films have been formed using low-cost methods such as spin-coating and spray-pyrolysis deposition in ambient air, making them a more appealing option. However, there is limited information on the compatibility of Cu_2O with organic/inorganic hybrid solar cells.[54,55]

This study offers a computational model (structural, electrical, and optical characteristics) for Pb-free hexagonal $Cs_3Bi_2I_9$ perovskite using well-known CASTEP software. The isotropic ion characteristic of Cs makes it a suitable option for materials sorting. We examine the band dispersion and bonding characteristics of the title substance, as well as its optical response qualities, based on the DFT calculations.[56−61] After that, SCAPS-1D computationally selected 49 different combinations of ETLs and HTLs to find the optimal combination for the $Cs_3Bi_2I_9$ absorber layer for the very first time. The main objective of this study was to identify ITO/ETL/$Cs_3Bi_2I_9$/HTL/Au, a highly efficient solar cell combination, to reduce the time and money needed to fabricate experimentally a wide variety of solar cell combinations. After selecting the most promising configurations from 49 combinations, we investigated the effect of the thickness of the $Cs_3Bi_2I_9$ absorber and the ETLs on device performance. Additionally, the effect of inherent resistances and capacitances, recombination, and generation rates on the PV performance is analyzed in detail. The results of our analysis are further validated using the current density−voltage (J−V) and quantum efficiency (QE) characteristics.

The organization of the paper is as follows: Section 1 gives a brief overview of the current state of the art of $Cs_3Bi_2I_9$ PSCs along with a summary of the work carried out in this research. Section 2 deals with the numerical methodology adopted within

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the framework of the DFT and SCAPS-1D simulator and the details of the basic PSC architecture. Section 3 elaborates on the structural and optoelectronic properties of the absorber along with a discussion on the PV performance of the optimized configurations of the PSC through a variation in different parameters. The comparison of our computed results with some of the existing theoretical and experimental results is also done in this section. Finally, Section 4 provides a brief summary of our research work.

2. MATERIALS AND METHODOLOGY

2.1. First Principle Calculations of the $Cs_3Bi_2I_9$ Absorber Layer. In the present work, the ground state geometry, electronic structure, and optical properties of the $Cs_3Bi_2I_9$ compound were numerically investigated with the CASTEP code. In order to solve the Kohn–Sham total energy equations, generalized gradient approximation (GGA) developed by Perdew–Burke–Ernzerhof (PBE) was considered as the exchange-correlation functional.[62,63] The used PBE-GGA is a common density functional in an electronic structure and optical calculations because it is computationally efficient and provides reasonable accuracy for many types of systems. Thus, for our studied system we used this potential as it produces reasonable and consistent electronic structure and optoelectronic properties in the DFT scheme of the CASTEP code. The electron–nucleus potential energy was treated with computationally economical ultrasoft pseudopotentials of the Vanderbilt.[64] Moreover, the projector-augmented wave (PAW) approach was used to explain the core–valence interaction.[65] For structural optimization in this study, the $Cs_3Bi_2I_9$ perovskite's hexagonal (space group $P6_3/mmc$) crystal structure was taken into consideration.[66] An energy cutoff of 500 eV for the basis sets was used for expansion. To determine the ground state energy, a quasi-Newton optimization technique, namely, the Broyden–Fletcher–Goldfarb–Shannon (BFGS) algorithm, was used.[67] The ground state total energy and lattice constant were obtained with a $10 \times 10 \times 4$ k-points mesh in the Brillouin zone of the Monkhorst–Pack (MP) scheme. The electronic properties including band structure, density of states (DOS), and optical properties were computed with these k-points, whereas a denser k-points mesh of $14 \times 14 \times 7$ size was employed to calculate the charge density map. The minimum energy difference between two consecutive iterations was taken as 10^{-4} eV/atom for convergence. The atoms were allowed to move under a maximum force of 0.03 eV/Å, displacement of 0.001 Å, and stress of 0.05 GPa during the relaxation of the atomic positions.

2.2. SCAPS-1D Numerical Simulation. The PV performance metrics of Pb-free PSCs (ITO/ETL/$Cs_3Bi_2I_9$/HTL/Au) were numerically simulated with SCAPS-1D software, which was developed by Prof. M. Burgelman.[68] The fundamental Poisson's and continuity equations are used as the starting equations in SCAPS-1D.[69−72] Poisson's equation can be written as

$$\nabla^2\psi = q/\varepsilon(n - p + N_A - N_D) \quad (1)$$

where $N_{D/A}$ denotes donor/acceptor concentration and ψ is the electrostatic potential. The equations of continuity can be illustrated as follows:

$$\nabla . J_n - q\partial n/\partial t = +qR \quad (2)$$

$$\nabla . J_p + q\partial p/\partial t = -qR \quad (3)$$

In eqs 2 and 3, J_n and J_p denote the electron and hole current densities, respectively, while R denotes the carrier recombination rate.

Drift and diffusion current relations used to obtain the total current density under the effect of an electric field and concentration gradient are as shown below:

$$J_n = qn\mu_n E + qD_n\nabla n \quad (4)$$

$$J_p = qp\mu_p E - qD_p\nabla p \quad (5)$$

Here, D_n and D_p denote the electron and hole diffusion coefficients, respectively.

2.3. $Cs_3Bi_2I_9$ Perovskite Solar Cell Structure. Figure 1 depicts the simulation setup for the inverted planar PSCs, which consists of

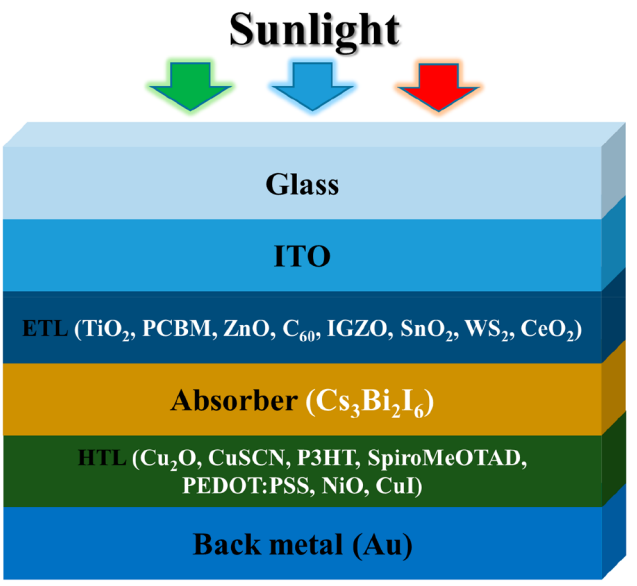


Figure 1. Configuration of $Cs_3Bi_2I_9$ solar cells used in this numerical simulation.

ITO/ETL/$Cs_3Bi_2I_9$/HTL/Au. The $Cs_3Bi_2I_9$ absorber is sandwiched between the *n*-type ETLs and the *p*-type HTLs enlisted in Tables 1 and 2, accordingly. The most important characteristics of the materials are included in Table 1 from various sources in the literature. Table 3 displays the defect states that are introduced at the $Cs_3Bi_2I_9$/ETL and $Cs_3Bi_2I_9$/HTL interfaces, respectively.

In this study, indium-doped tin oxide (ITO), $Cs_3Bi_2I_9$, and gold (Au) are used as the transparent conductive oxide (TCO), absorber, and back contact, respectively, with seven different ETLs and seven HTLs for SCAPS-1D simulations to analyze the different PSC architectures and determine the best pairing (Tables 1 and 2). Simulations were carried on at an ambient temperature of 300 K under the AM1.5G 1 solar spectrum with an incoming power density and frequency of 100 mW/cm^2 and 1 MHz, respectively.

3. RESULTS AND DISCUSSION

3.1. Analysis of DFT Results. *3.1.1. Structural Properties of $Cs_3Bi_2I_9$ Compound.* The semiconducting material $Cs_3Bi_2I_9$ usually belongs to two types of crystal systems, namely, hexagonal and monoclinic symmetry. However, the $Cs_3Bi_2I_9$ halide perovskite forms a hexagonal crystal structure (space group $P6_3/mmc$, a = 8.411 Å, and c = 21.18 Å).[74−76] In the present study, the hexagonal $Cs_3Bi_2I_9$ crystalline structure in the $P6_3/mmc$ space group is considered for the entire calculations to achieve the ground state minimum energy. The best-optimized crystal lattice parameters of the material system are $a = b$ = 8.5785 Å and c = 22.0245 Å, with a current cell volume of 1403.6878 Å^3. The calculated lattice parameters are consistent with the experimental data as well as previous reports,[21,66,76] which indicates the structural stability of the compound. The estimated formation energy of the title material was −1.315 eV/atom, which also confirms its stability. Figure 2 depicts the completely relaxed hexagonal structure of $Cs_3Bi_2I_9$ perovskite with the lowest total energy. The structure consists of a total of 28 atoms, two topologically Cs^{1+} sites, and 12 I^{1-} ions bound to both Cs^{1+} sites to form CsI_{12} cuboctahedra. They exchange

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Table 1. Input Parameters for TCO, ETL, and Absorber Layer ($Cs_3Bi_2I_9$)[25,73]

Parameters	ITO	TiO_2	PCBM	ZnO	C_{60}	IGZO	SnO_2	WS_2	$Cs_3Bi_2I_9$
Thickness (nm)	500	30	50	50	50	30	100	100	800
Band gap, E_g (eV)	3.5	3.2	2	3.3	1.7	3.05	3.6	1.8	2.03
Electron affinity, X (eV)	4	4	3.9	4	3.9	4.16	4	3.95	3.55
Dielectric permittivity (relative), ε_r	9	9	3.9	9	4.2	10	9	13.6	9.68
CB effective density of states, N_C (1/cm^3)	2.2×10^{18}	2×10^{18}	2.5×10^{21}	3.7×10^{18}	8.0×10^{19}	5×10^{18}	2.2×10^{18}	1×10^{18}	4.98×10^{19}
VB effective density of states, N_V (1/cm^3)	1.8×10^{19}	1.8×10^{19}	2.5×10^{21}	1.8×10^{19}	8.0×10^{19}	5×10^{18}	1.8×10^{19}	2.4×10^{19}	2.11×10^{19}
Electron mobility, μ_n (cm^2/(V s))	20	20	0.2	100	8.0×10^{-2}	15	100	100	4.3
Hole mobility, μ_h (cm^2/(V s))	10	10	0.2	25	3.5×10^{-3}	0.1	25	100	1.7
Shallow uniform acceptor density, N_A (1/cm^3)	0	0	0	0	0	0	0	0	1×10^9
Shallow uniform donor density, N_D (1/cm^3)	1×10^{21}	9×10^{16}	2.93×10^{17}	1×10^{18}	1×10^{17}	1×10^{17}	1×10^{17}	1×10^{18}	1×10^9
Defect density, N_t (1/cm^3)	1×10^{15}[a]	1×10^{15}[a]	1×10^{15}[a]	1×10^{15}[a]	1×10^{15}[a]	1×10^{15}[a]	1×10^{15}[a]	1×10^{15}[a]	1×10^{15}

[a]The values in the table remain constant during the initial optimization procedure to get the best possible combination of HTL, ETL, and absorber.

Table 2. Input Parameters for Different HTLs[73]

Parameters	Cu_2O	CuSCN	P3HT	PEDOT:PSS	Spiro-MeOTAD	NiO	CuI
Thickness (nm)	50	50	50	50	200	100	100
E_g (eV)	2.2	3.6	1.7	1.6	3	3.8	3.1
X (eV)	3.4	1.7	3.5	3.4	2.2	1.46	2.1
ε_r	7.5	10	3	3	3	10.7	6.5
N_C (1/cm^3)	2×10^{19}	2.2×10^{19}	2×10^{21}	2.2×10^{18}	2.2×10^{18}	2.8×10^{19}	2.8×10^{19}
N_V (1/cm^3)	1×10^{19}	1.8×10^{18}	2×10^{21}	1.8×10^{19}	1.8×10^{19}	1×10^{19}	1×10^{19}
μ_n (cm^2/(V s))	200	100	1.8×10^{-3}	4.5×10^{-2}	2.1×10^{-3}	12	100
μ_h (cm^2/(V s))	8600	25	1.86×10^{-2}	4.5×10^{-2}	2.16×10^{-3}	2.8	43.9
N_A (1/cm^3)	1×10^{18}	1×10^{18}	1×10^{18}	1×10^{18}	1.0×10^{18}	1×10^{18}	1.0×10^{18}
N_D (1/cm^3)	0	0	0	0	0	0	0
N_t (1/cm^3)	1.0×10^{15}[a]	1×10^{15}[a]	1×10^{15}[a]	1×10^{15}[a]	1.0×10^{15}[a]	1×10^{15}[a]	1.0×10^{15}[a]

[a]The values in the table remain constant during the initial optimization.

Table 3. Input Parameters for Interface Defect Layers[73]

Interface	Defect type	Capture cross section: electrons/holes (cm^2)	Energetic distribution	Reference for defect energy level	Total density (cm^{-2}) (integrated over all energies)
ETL/$Cs_3Bi_2I_9$	Neutral	1.0×10^{-17} 1.0×10^{-18}	Single	Above the VB maximum	1.0×10^{10}
$Cs_3Bi_2I_9$/HTL	Neutral	1.0×10^{-18} 1.0×10^{-19}	Single	Above the VB maximum	$\times 10^{10}$

corners with three equivalent BiI_6 octahedra and nine equivalent CsI_{12} cuboctahedra. The constructed substructure also has four comparable BiI_6 octahedra and seven CsI_{12} cuboctahedra on its faces, where the octahedral edge tilt angles are approximately 19°. Cs−I bond lengths vary and range from 4.22 to 4.49 Å. Six Cs−I bonding lengths are smaller (4.27 Å) and six are larger (4.37 Å). Three comparable CsI_{12} cuboctahedra share corners with the Bi^{3+}-bound six I^{1-} atoms that make up the BiI_6 octahedron. Additionally, three Bi−I bond lengths are shorter (2.96 Å) and three are longer (3.23 Å). The first I^{1-} site is coupled to four Cs^{1+} and one Bi^{3+} atom in a 1-coordinate geometry, resulting in two inequivalent I^{1-} sites. I^{1-} is bound to four Cs^{1+} and two equivalent Bi^{3+} atoms in a 2-coordinate geometry at the second I^{1-} site to produce the whole structural layout.

3.1.2. Band Structure and Density of States of $Cs_3Bi_2I_9$ Compound. In order to explore and understand the electronic behavior of $Cs_3Bi_2I_9$, an energy band diagram (EBD) is shown in Figure 3(a), obtained by calculating the energy of the electronic states as a function of k (wave vector in reciprocal lattice). This $E-k$ diagram provides direct information about the electronic properties of the material, such as metallic, semiconducting, half-metallic, or insulating behavior, nature, and magnitude of the band gap (if it exists), and carrier mobility of the material. It is clear from the figure that, at Fermi level E_F (0 eV energy), there is no overlap of the occupied electronic states (called the valence band) with unoccupied electronic states (called the conduction band), due to which a distinct energy gap exists around the E_F, and therefore the material is said to be a semiconductor. The nature of the band gap (E_g) is confirmed by locating the position of the valence band maxima (VBM) and conduction band minima (CBM) with respect to the k-vector value. It is noticed that VBM (CBM) is located at the high symmetry K-point (G-point) in the Brillouin zone; thus, VBM and CBM are located at two different k-values, an obvious characteristic of the presence of an indirect band gap. The magnitude of the band gap is 2.453

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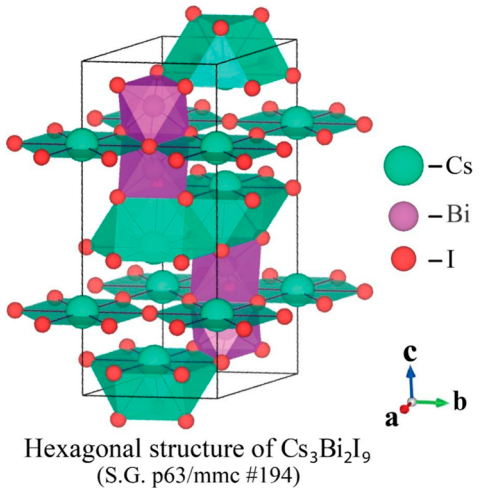


Figure 2. Image illustrating the $Cs_3Bi_2I_9$ hexagonal crystal structure. Black lines designate the unit cell dimensions, and purple and olive-green colors indicate the bismuth and cesium iodide octahedra, respectively.

eV, which is the energy difference between the VBM (here it is located at 0 eV) and the CBM (here it is positioned at 2.453 eV above the E_F). This indirect band gap value is in agreement with the literature[76,77] and might be suitable for PV applications. The formation and distribution of energy bands are further examined with the density of states (DOS) plots shown in Figure 3(b). It is shown that electronic states in the valence band edge are mainly contributed by the I 5p and Bi 6s electrons, while the Bi 6p and Cs 6s/5p electrons have a very small contribution, and the I 5s electrons have almost no contribution. On the other hand, Cs 6s/5p, Bi 6s/6p, and I 6s/5p conduction electrons contribute to the electronic states in the conduction band bottom. The collective presence of Cs, Bi, and I electronic states in the region below the E_F shows the strong hybridization among the s and p states of all atoms, which results in the strong binding between the atoms.

3.1.3. Charge Density of the $Cs_3Bi_2I_9$ Compound. The study of charge density mapping is essential for a comprehensive explanation of the charge transfer and kind of interaction among the constituent atoms of the $Cs_3Bi_2I_9$ hexagonal perovskite. Figure 3(c) depicts the charge density map along the (111) crystal plane (2D) to show the atomic charge distribution clearly. The scale on the right side displays the magnitude of electron density, with blue and red hues signifying the greatest and smallest intensities of electron density, correspondingly. It is seen that around the I atom, the charge density is shown to be at its maximum, whereas around the Bi atom, it is at its lowest value. Because the charge contours of the Cs and I elements do not coincide, ionic bonding is present between these two atoms, whereas covalent bonding between the Bi and I atoms is clearly visible in the spherical shape of their charges. The hybridization between the Bi 6s and I 5s orbitals, which is also evident from the DOS curve, gives potency to the Bi−I bond. Furthermore, the electronic charge density maps along several other crystal planes yielded similar findings, indicating that the behavior is isotropic

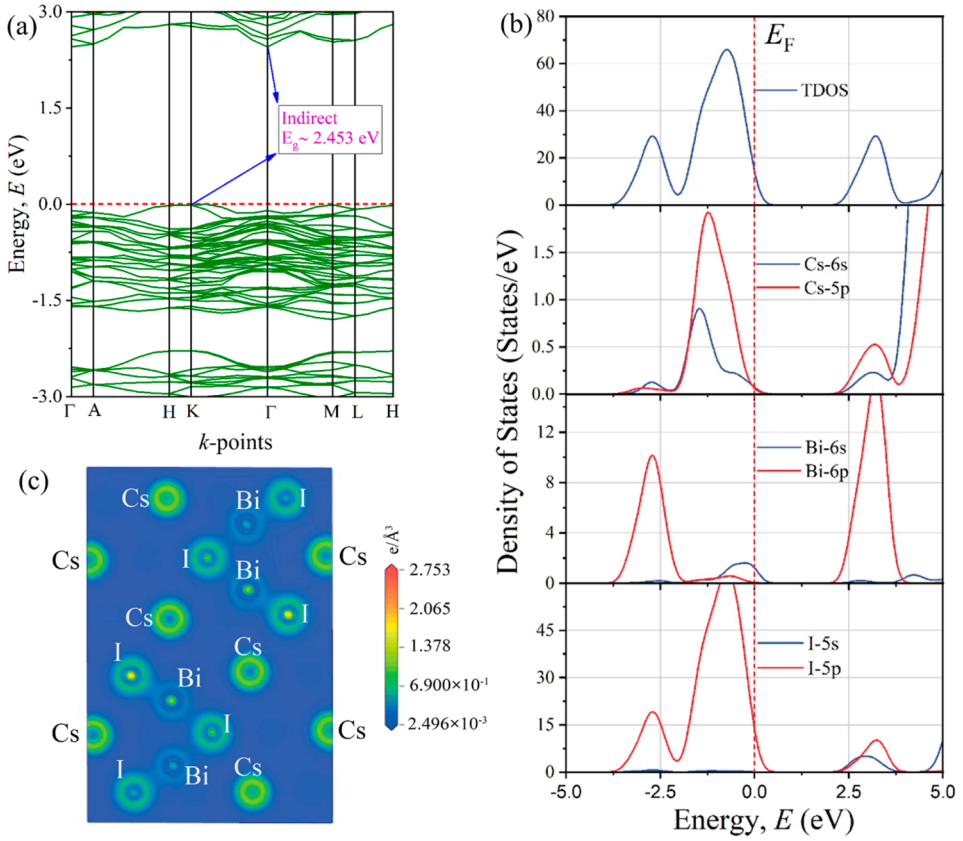


Figure 3. (a) Electronic band structure, (b) density of states, and (c) electronic charge density contour of $Cs_3Bi_2I_9$ halide perovskite along the (111) plane.

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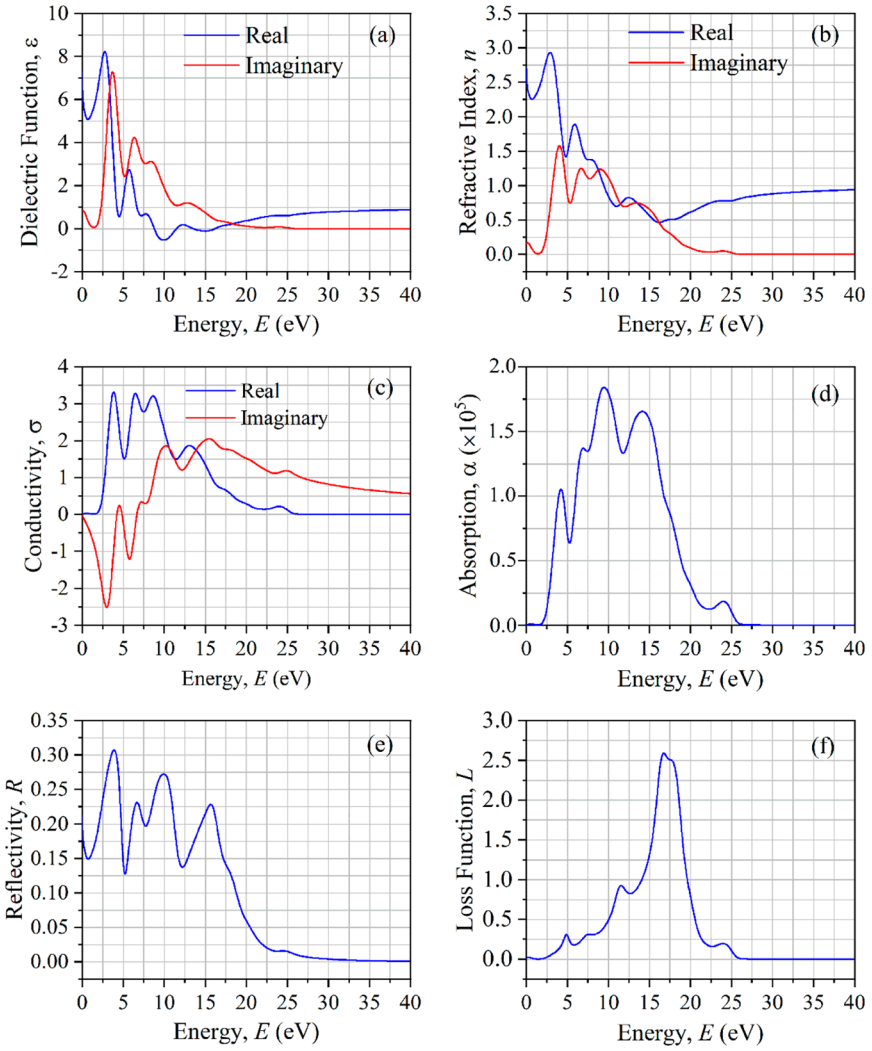


Figure 4. Computed optical properties of the $Cs_3Bi_2I_9$ halide perovskite.

and the charge distribution is substantially round throughout the entire atoms as well.[73,78]

3.1.4. Optical Properties of the $Cs_3Bi_2I_9$ Compound. The importance of the forbidden energy gap cannot be neglected in view of the optimization, expansion, and improvement in the applications of semiconductors. The photon harvesting capability of an optoelectronic device is largely dependent on the bandgap of the absorber material.[79] The optical properties of the semiconducting materials can be tuned depending on the field of its application, due to which these properties have gained the noticeable attention of researchers. This subsection addresses the optical properties and relevant applications of $Cs_3Bi_2I_9$. Various optical parameters such as complex dielectric function, reflectivity, optical conductivity, refractive index, and absorption coefficient are used to analyze the different optical responses of any medium. A complex dielectric function provides a detailed picture of the optical response of materials when these materials come in contact with the electromagnetic (EM) radiation. The interaction between an electron and a phonon can also be defined by a complex dielectric function. A complex dielectric function has a real part that is denoted by $\varepsilon_1(\omega)$, and this real part is used to examine the energy conducted by the medium. Energy stored in a medium can be taken out by applying Kramers−Kronig transformation by imposing the imaginary part $\varepsilon_2(\omega)$ of the dielectric function.[80] Further, the explanation of band structure, absorption pattern, and shifting of charge carriers from VB to CB can also be done by using $\varepsilon_2(\omega)$.[81−83]

The complex dielectric function measures the response of the material against the surrounding electric and magnetic fields. Equation 6 gives the mathematical form of the complex dielectric function.

$$\varepsilon = \varepsilon_1(\omega) + \varepsilon_2(\omega) \tag{6}$$

The variation in the values of $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$ of investigated compound $Cs_3Bi_2I_9$ is shown in Figure 3. The frequency at which a material transforms from conductor to insulator is known as the zero frequency or plasma frequency, and its perfect measurement is of huge importance. The zero frequency limit $\varepsilon_1(0)$ strongly relies on band gap, and it is the electronic ingredient of the static dielectric constant. From the figure, the value of zero frequency limit, $\varepsilon_1(0)$, was found to be approximately 7.1. $\varepsilon_1(\omega)$ has positive and negative halves, as shown in Figure 4(a). The positive half is for propagation, while the negative half is for absorption of EM waves. After attaining a peak value of 8.1 at 3 eV, $\varepsilon_1(\omega)$ shows a sharp decline in the spectra, and it becomes negative at about 9.7 eV. The frequency at which these curves start to turn negative is called the plasmon

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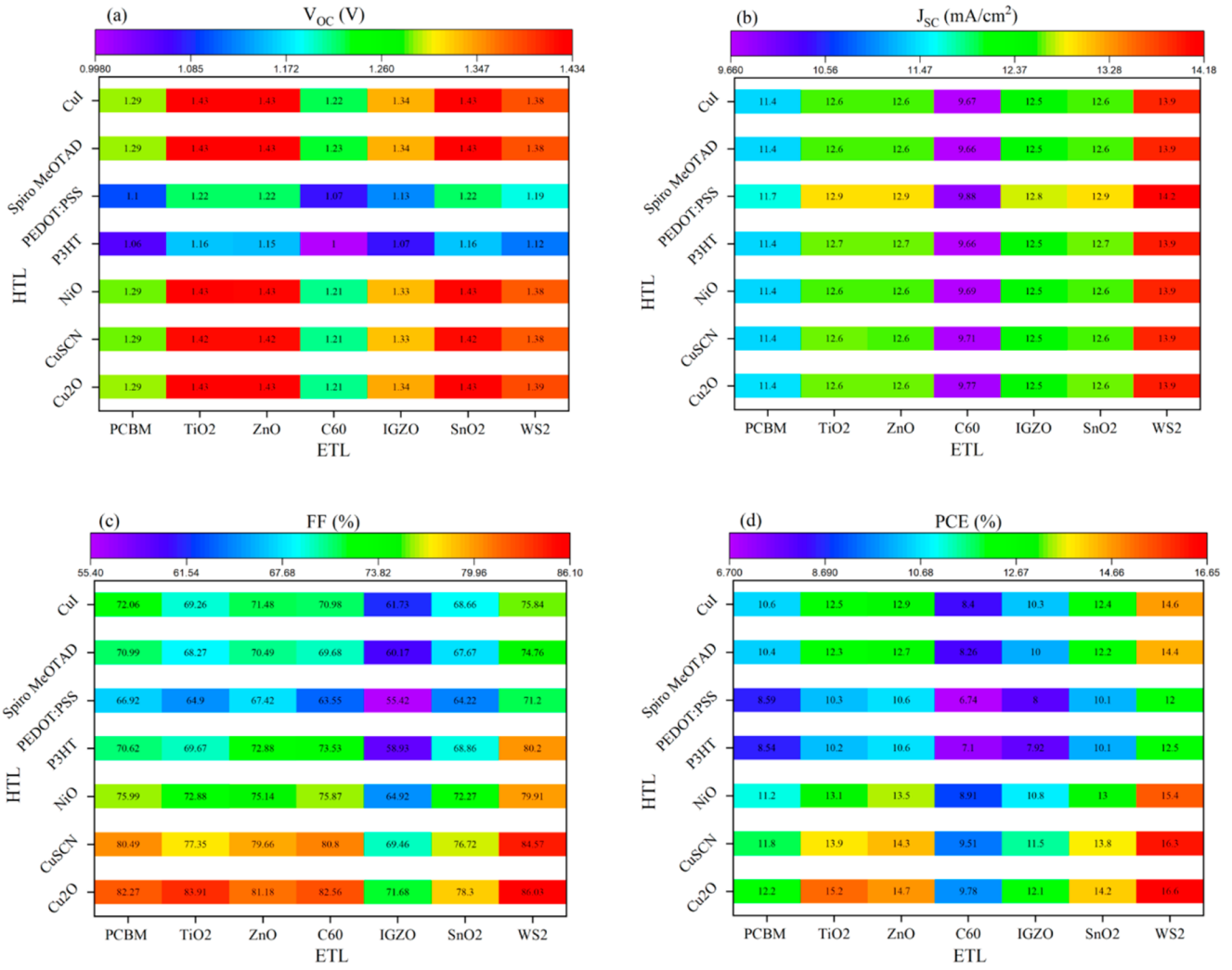


Figure 5. Impact for different HTLs and ETLs on PV characteristics: (a) V_{OC} (V), (b) J_{SC} (mA/cm^2), (c) *FF* (%), and (d) PCE (%) of $Cs_3Bi_2I_9$-based PSC while Au is used as the back contact.

frequency (ω_p). The figure shows that the positive section of ε_1 has a peak value in the minor energy series, indicating its conducting nature, while for the higher energy range $Cs_3Bi_2I_9$ acts as an insulator. $\varepsilon_1(\omega)$ has a peak value in the visible energy range, which represents that this compound is a suitable contender for optoelectronic devices. $\varepsilon_2(\omega)$ is associated with the photon's absorption and determines the electronic properties of crystalline materials. Figure 4(a) shows that the spectra of $\varepsilon_2(\omega)$ have a peak value at around 4.9 eV. The peaks highlight the movement of charge carriers from occupied to unoccupied bands. For higher energy ranges, $\varepsilon_2(\omega)$ starts decreasing and becomes zero at approximately 20 eV. Further, the absorption spectrum of the proclaimed compound indicates a negligible absorption in the visible energy region, which further supports the results of $\varepsilon_1(\omega)$.

On the other hand, the frequency-dependent refractive index assists in explaining the dissemination of light in any medium. Furthermore, the complex refractive index is adequate to describe the material's optical properties. The change in velocity or wavelength of an electromagnetic (EM) wave as it travels from a vacuum into a medium is described by the real part of the refractive index, while the dissipation rate of the wave within the medium is explained by its imaginary part. Figure 3(b) displays the $n(\omega)$ vs energy plot of $Cs_3Bi_2I_9$. Zero frequency values of $n(\omega)$ are referred to as static values of refractive index $n(0)$. The relation between $\varepsilon_1(0)$ and $n(0)$ is given by the following expression (eq 7):

$$n(0) = \sqrt{\varepsilon_1(0)} \tag{7}$$

Figure 4(b) shows that the static values of $n(0)$ for the real and imaginary parts of the investigated compound are approximately 2.7 and 0.29, respectively. After a slight decrement of static value peaks of the real and imaginary parts, $n(\omega)$ starts increasing and attains its uppermost values of 2.8 and 1.65 at energies of 4.6 and 4.8 eV, respectively, which signifies the utmost interaction between the photon and the medium.

Optical conductivity is the ability of any medium to initiate a conduction process as the radiations interact with it. Figure 4(c) discloses the behavior of the real and imaginary parts of optical conductivity. The figure shows that the real optical conductivity starts rising from 2 eV and attains a maximum value of 3.4 at 4.9 eV. However, the figure shows that $Cs_3Bi_2I_9$ becomes an insulator for higher energy ranges, and the real part approaches zero at about 25 eV. The imaginary optical conductivity exhibits a reverse trend from zero frequency to approximately 6.4 eV in contrast to the real part.

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An elementary optical parameter that can measure the quantity of energy absorbed by any medium is known as the absorption coefficient $\alpha(\omega)$. The absorption spectra enumerate the fraction of light intensity attenuation with distance. Figure 4(d) depicts the $\alpha(\omega)$ versus energy plot. It can be observed from the plot that there is no sharp peak of $\alpha(\omega)$ in IR as well as in the visible energy range, which indicates that $Cs_3Bi_2I_9$ has the capability to absorb both visible and IR rays. This behavior of optical absorption can be utilized for applications in solar PV systems. However, peaks become more prominent in the UV region, which signifies the phenomenon of blue shift.

The parameter reflectivity determines whether a substance has the properties of an antireflecting coating or not. Figure 4(e) reveals the reflectivity of $Cs_3Bi_2I_9$ within the energy range of 0 to 40 eV, where it can be noticed that the static level value of R is 0.22. Further, R starts growing and achieves its crest value beyond the visible region and becomes zero at around 30 eV.

Figure 4(f) presents the energy loss function (L), which is another important optical parameter used to quantify the loss of energy of fast-moving electrons in a medium. The figure reveals the negligible loss of energy of electrons in the IR and visible regions. The figure shows that the highest energy loss occurs in the UV region at about 15.2 eV.

3.2. Analysis of SCAPS-1D Results. Here, we evaluated the PV performance of the ITO/ETL/$Cs_3Bi_2I_9$/HTL/Au by changing the thicknesses of the ETL, $Cs_3Bi_2I_9$ absorber, and HTL layers while keeping the acceptor concentration and defect densities constant at 10^{18} and 10^{15} cm^{-3}, respectively. After the initial simulation, the best-performing devices were analyzed based on the results of 49 possible combinations to determine the following behaviors (Figure 5 and Table 4).

Table 4. Performance Parameters for Best Device Configuration for Each of the ETLs

Optimized device	V_{OC} (V)	J_{SC} (mA/cm^2)	*FF* (%)	PCE (%)
ITO/PCBM/$Cs_3Bi_2I_9$/Cu_2O/Au	1.29	11.44	82.27	12.16
ITO/TiO_2/$Cs_3Bi_2I_9$/Cu_2O/Au	1.43	12.65	83.91	15.17
ITO/ZnO/$Cs_3Bi_2I_9$/Cu_2O/Au	1.43	12.64	81.18	14.70
ITO/C_{60}/$Cs_3Bi_2I_9$/Cu_2O/Au	1.21	9.77	82.56	9.78
ITO/IGZO/$Cs_3Bi_2I_9$/Cu_2O/Au	1.34	12.54	71.68	12.06
ITO/SnO_2/$Cs_3Bi_2I_9$/Cu_2O/Au	1.43	12.64	78.30	14.18
ITO/WS_2/$Cs_3Bi_2I_9$/Cu_2O/Au	1.39	13.93	86.03	16.61

3.2.1. Impact of ETL and HTL Variation. Our study is focused on a PSC made from $Cs_3Bi_2I_9$, with an initial configuration of around seven ETLs, seven HTLs, and Au as the back contact. Six sets of optimal solar cells are reported after numerically testing 49 different configurations (Figures 5(a–d)) of SCs in the SCAPS-1D platform.

As can be seen in Table 4, the 49 configurations whose results are reported in Figure 5 were narrowed down to just six using the criteria of optimal performance. In this study, we employed a total of seven HTLs, namely, Cu_2O, CuSCN, NiO, P3HT, PEDOT:PSS, Spiro-MeOTAD, and CuI, with seven ETLs, namely, PCBM, TiO_2, ZnO, C_{60}, IGZO, SnO_2, and WS_2, to compare the PV performance of 49 configurations. All of the ETLs, except for C_{60}, performed well with Cu_2O HTL by delivering PCE ∼12−17%, as shown in Figure 5(d). While assessing the performance, the $Cs_3Bi_2I_9$ absorber appears to be best suited with HTLs like Cu_2O, CuSCN, NiO, Spiro-MeOTAD, and CuI (Figure 5). When compared to the other HTLs, the efficacy of P3HT and PEDOT:PSS was diminished when combined with any ETL. As a p-type layer, the HTL needs to be thinner than the n-type ETL to prevent recombination and to facilitate the instantaneous exchange of a sufficient number of charge carriers with the structure. As mentioned earlier, Cu_2O has been thought to be a feasible HTL due to its p-type semiconducting properties along with its direct bandgap nature coupled with its nontoxicity, compatibility with different organic/inorganic SCs, and availability on the earth's surface.[54,55] The highest throughput was seen with Cu_2O as the HTL and PCBM, TiO_2, ZnO, C_{60}, IGZO, SnO_2, and WS_2 as ETLs, as shown in Figure 5.

Figure 5(a) depicts that TiO_2, ZnO, and SnO_2 ETLs showed the highest open circuit voltage (V_{OC}) of ∼1.43 eV, while C_{60} ETL showed the lowest V_{OC} of 1 V for the respective HTL. The short circuit current (J_{SC}) of the WS_2 ETL was the highest in the range 13.9−14.2 mA/cm^2 among all of the studied ETLs, as illustrated in Figure 5(b). TiO_2, ZnO, IGZO, and SnO_2 ETLs showed similar sorts of performance with J_{SC} in the range of 12.6−12.9 mA/cm^2, and the C_{60} ETL showed the lowest range of J_{SC} (9.66−9.89 mA/cm^2). Figure 5(c) shows that the highest fill factor (*FF*) of 86.03% is exhibited by the WS_2 ETL, and the IGZO ETL showed the lowest *FF* of 55.42% among all of the studied transport layers. ETLs such as TiO_2 and WS_2 produced a comparatively greater efficiency of >15% among the studied ITO/ETL/$Cs_3Bi_2I_9$/Cu_2O/Au configurations. However, the performances of PCBM, ZnO, IGZO, and SnO_2 were also notable, and their efficiencies were 12.20%, 14.70%, 12.10%, and 14.20%, while ETL C_{60} showed the lowest efficiency of 9.78% (Figure 5(d)).

3.2.2. Band Structure of $Cs_3Bi_2I_9$ Configurations with Different ETLs. The band alignment of several heterostructures based on $Cs_3Bi_2I_9$ absorbers is depicted in Figures 6(a–f), which illustrates the minima and maxima of the conduction and valence bands, respectively. In all of the ETLs, F_p is in phase with E_V, and the harmonic relationship between F_n and E_C persists. Because TiO_2 and ZnO ETLs have almost the same bandgap compared to the other ETLs (Figures 6(b) and (c)), their performances with the same heterostructure are quite close to each other. But the bandgap of WS_2 ETL showed better transportation of electrons and holes in the interface layers (Figure 6(f)). Similarly, for the rest of the ETLs, this kind of action was not possible due to their band gaps and band alignments. While the holes are blocked by the ETL, the electron movement is prevented by the HTL. Holes are collected by the Au back contact from the Cu_2O HTL, whereas electrons are effectively collected by the ITO front contact. Specifically, a 5.1 eV work function (WF) for gold was evaluated for the back contact, whereas a 4.0 eV WF for indium dioxide was considered for the front contact.

3.2.3. Effect of Absorber and ETL Thickness on PV Performance. Figures 7−10 show the computed V_{OC}, J_{SC}, *FF*, and PCE contour maps for $Cs_3Bi_2I_9$-based PSCs as a function of ETL thickness (30−500 nm) and absorber thickness (300−1300 nm). The maximum V_{OC} of 1.30−1.471 V was observed at an absorber thickness of 300−400 nm with an ETL thickness ranging from 30 to 100 nm for all studied devices. As shown in Figure 7(b), the highest value of V_{OC} seen in any device was 1.472 V for a $Cs_3Bi_2I_9$ perovskite absorber with a thickness of

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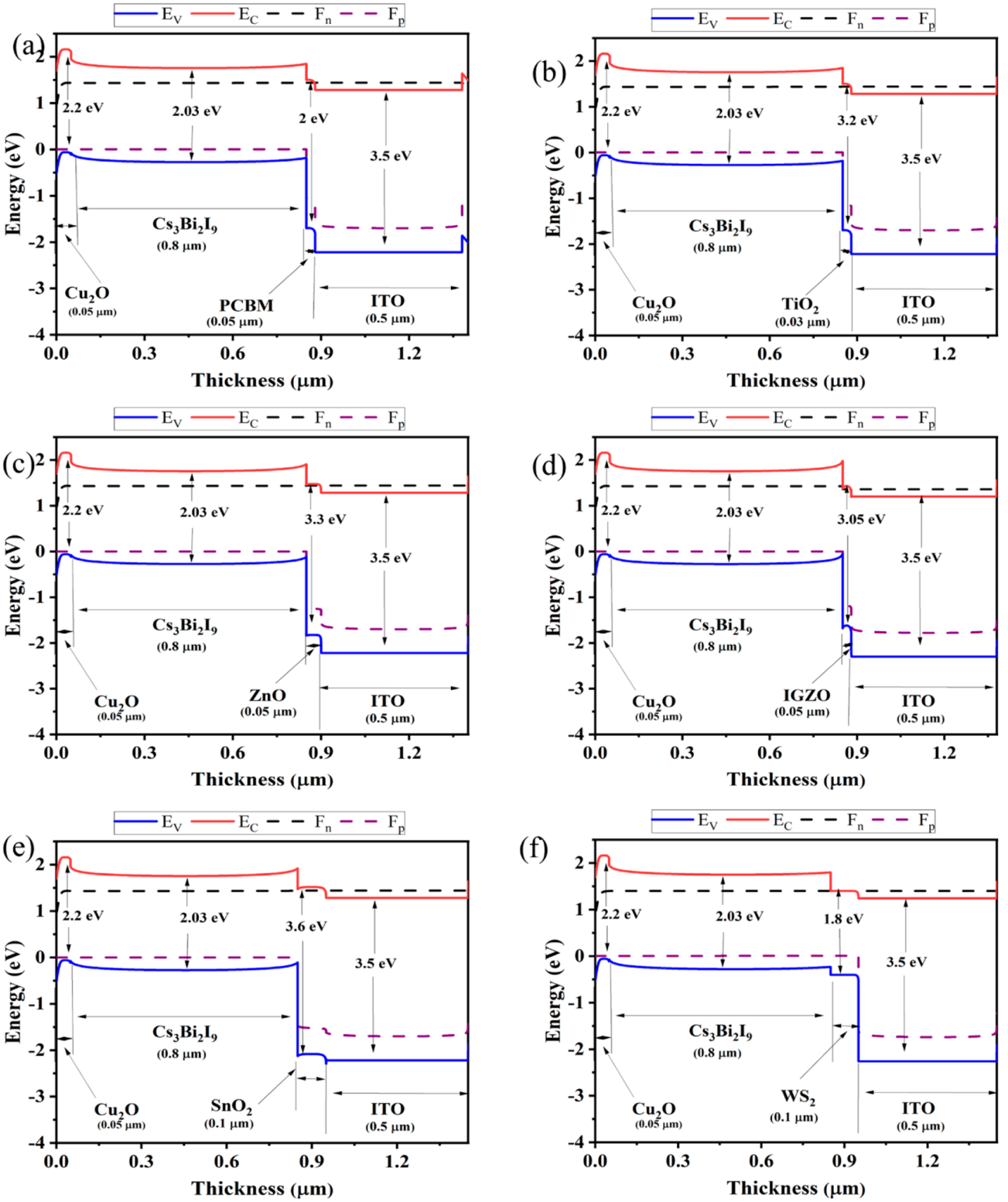


Figure 6. Band diagram of $Cs_3Bi_2I_9$-based PSCs for different HTLs with Au as the back contact and ETLs like (a) PCBM, (b) TiO_2, (c) ZnO, (d) IGZO, (e) SnO_2, and (f) WS_2.

400 nm and a TiO_2 ETL with a thickness of 50 nm. The V_{OC} values of the 400 nm thick absorber and 50 nm thick ZnO and SnO_2 ETLs were slightly lower than 1.470 V (Figures 7(c) and (d)). Among all the studied PSCs, the PCBM ETL showed a minimum V_{OC} of 1.30 V, while the absorber and ETL thicknesses were 400 and 100 nm, respectively (Figure 7(a)). Since the electron affinities of the six selected ETLs are all within a narrow range of 3.9−4.16 eV, the V_{OC} of each device is remarkably consistent at around 1.30−1.472 V. As can be seen in Figure 7, however, as the absorber and ETL thicknesses are increased over 500 and 50 nm, respectively, the V_{OC} values of all of the $Cs_3Bi_2I_9$-based PSCs start to drop.

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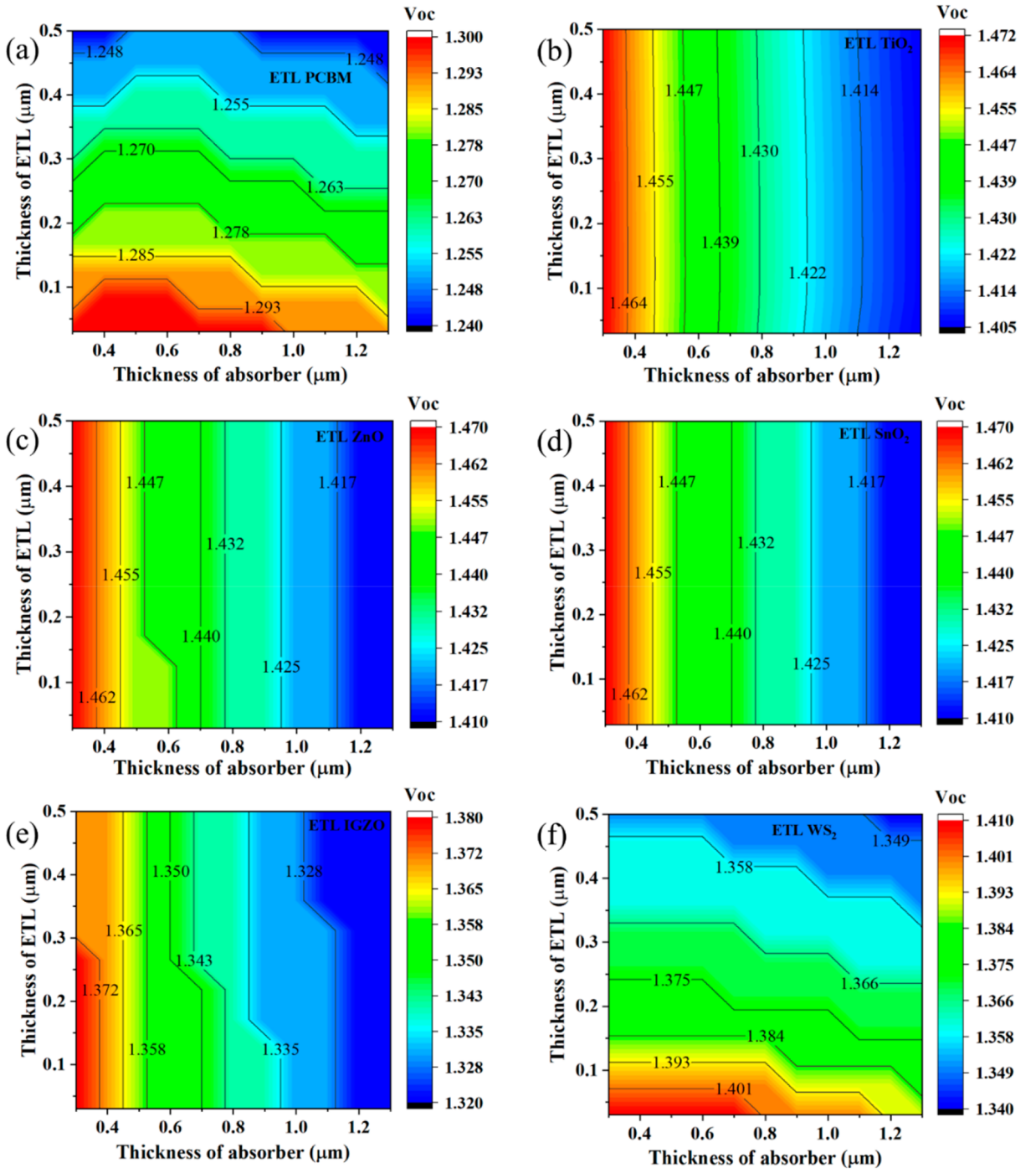


Figure 7. Contour plots of V_{OC} with the simultaneous variation of the thickness of $Cs_3Bi_2I_9$ absorber and ETLs like (a) PCBM, (b) TiO_2, (c) ZnO, (d) SnO_2, (e) IGZO, and (f) WS_2.

The maximum J_{SC} was measured between 12.80 and 14.40 mA/cm^2 with an absorber thickness between 1000 and 1300 nm and an ETL thickness between 50 and 200 nm, as shown in Figure 8. Figure 8(a) illustrates the PCBM ETL with a thickness of 50 nm and an absorber thickness of 1200 nm that achieved the minimum J_{SC} of 12.80 mA/cm^2 among the six investigated configurations. In contrast, the WS_2 ETL with absorber and ETL thicknesses of 1200 and 150 nm, respectively, delivered a J_{SC} of a maximum of 14.40 mA/cm^2, which was the best among the studied structures (Figure 8(f)). Because of the larger band gap present in these studied ETLs, a greater fraction of the incident light can be absorbed by the $Cs_3Bi_2I_9$ absorber layer in configurations constructed of TiO_2 and WS_2. Photons of longer wavelengths are absorbed when the absorber is thicker, resulting in a better J_{SC}. Below an absorber thickness of 1000 nm and an ETL thickness of over 200 nm, however, the J_{SC} of all devices begins to drop due to the dominance of recombination over the carrier lifetime in the devices.

Figure 9 displays the *FF* values for six ETL configurations for all of the selected and prospective structures, with the maximum *FF* in the range of 78.10−89.10%, with the absorber and ETL thicknesses varying from 300 to 1300 nm and 30 to 500 nm,

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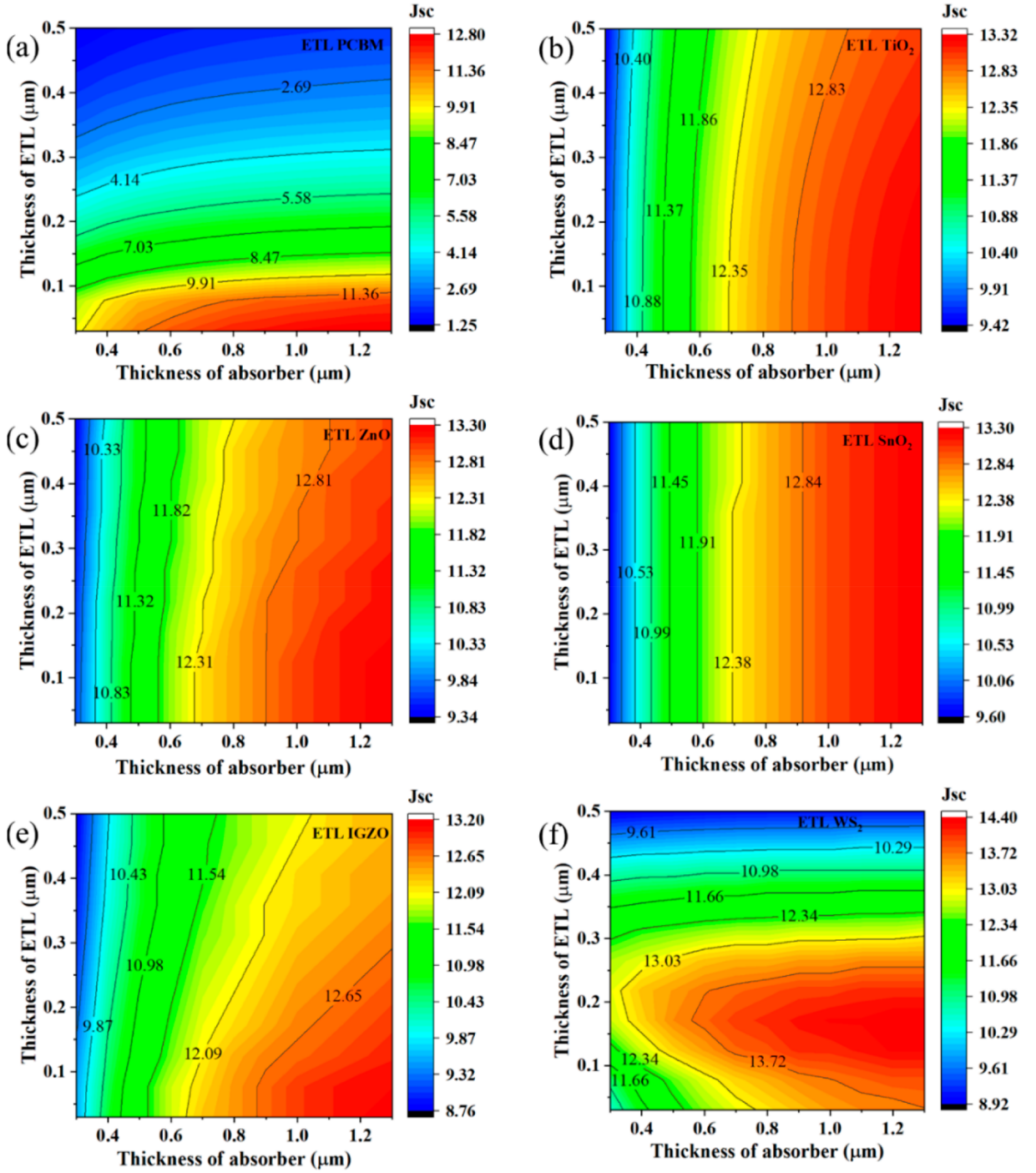


Figure 8. Contour plots of J_{SC} with a simultaneous variation of the thickness of the $Cs_3Bi_2I_9$ absorber and ETLs like (a) PCBM, (b) TiO_2, (c) ZnO, (d) SnO_2, (e) IGZO, and (f) WS_2.

respectively. The WS_2 ETL with absorber and ETL thicknesses of 400 and 100 nm, respectively, had the highest *FF* of 89.10% (Figure 9(f)), while the IGZO ETL with absorber and ETL thicknesses of 300 and 50 nm, respectively, had an *FF* that was seen to be comparatively lowest at 78.10% in comparison to the other ETLs (Figure 9(e)). Extending the absorber thickness beyond the diffusion length caused a noticeable recombination of the photogenerated carrier in the quasi-neutral zone, resulting in a drop in the *FF* at absorber thicknesses of ≥400 nm and ETL thicknesses of ≥200 nm for all of the structures studied so far. Since the *FF* is responsible for the generation of a strong electric field across the absorber layers, increasing the thickness of the absorber layer results in a reduction in the strength of the electric field and the establishment of a quasi-neutral zone within the absorber layer.

With an ETL thickness of 50−200 nm and an absorber thickness of 600−1200 nm, the best PCE of 12.10−16.70% was recorded for all investigated $Cs_3Bi_2I_9$-based structures (Figure 10). Figure 10(f) depicted that the WS_2 ETL, with absorber and ETL thicknesses of 1000 and 200 nm, respectively, had a maximum PCE of 16.70%, as predicted by the PV parameters J_{SC} and *FF* findings discussed earlier. With absorber and ETL

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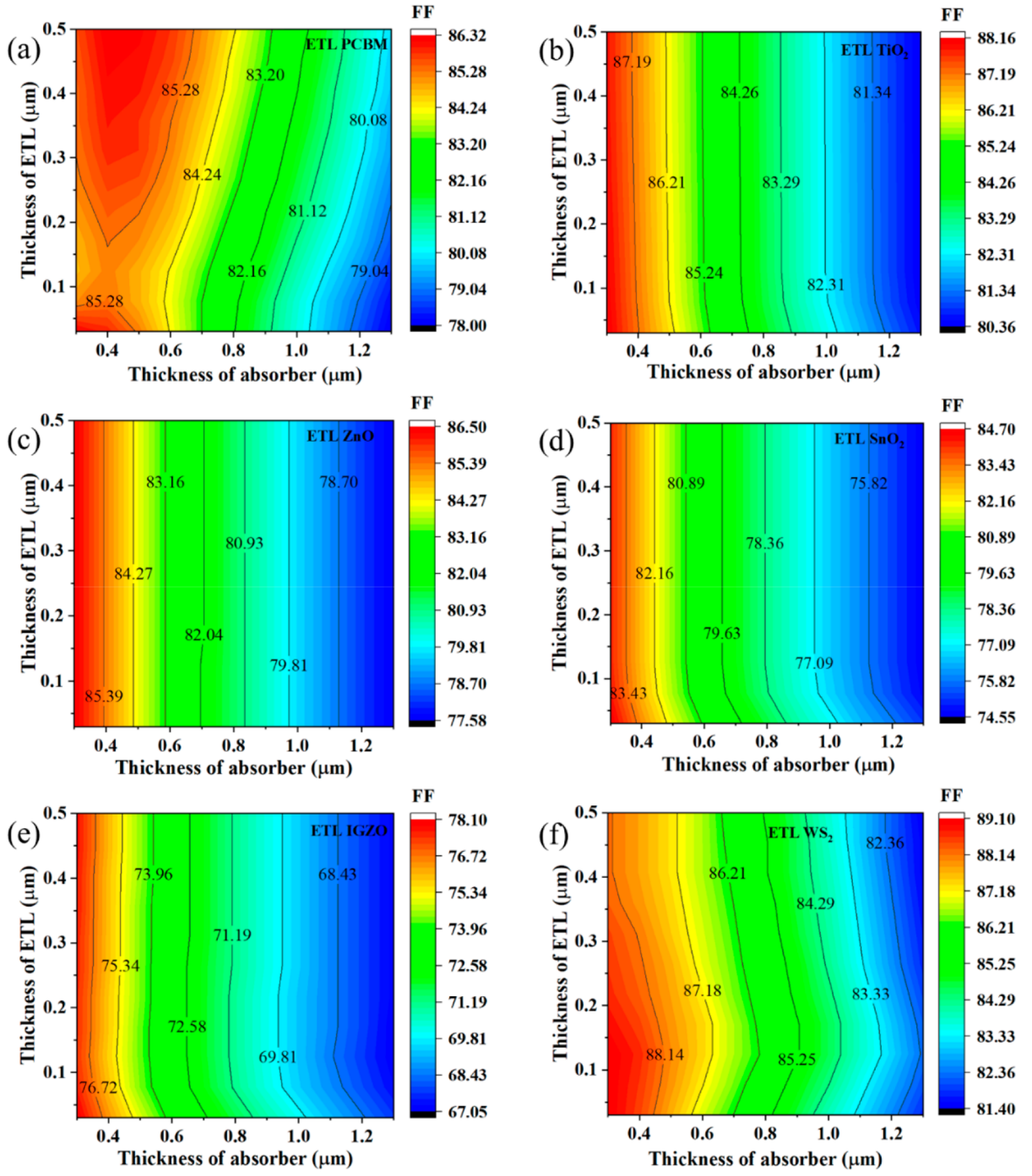


Figure 9. Contour plots of *FF* with a simultaneous variation of the thickness of $Cs_3Bi_2I_9$ absorber and ETLs like (a) PCBM, (b) TiO_2, (c) ZnO, (d) SnO_2, (e) IGZO, and (f) WS_2.

thicknesses of 800 and 50 nm, respectively, the IGZO ETL efficiency was determined to be relatively low at 12.10% (Figure 10(e)). As a result, for six distinct ETL configurations, the optimal thicknesses for the absorber and ETLs were determined to be 600−1200 and 50−200 nm, respectively. When compared to the other ETLs, however, the TiO_2 and WS_2-based structures that use $Cs_3Bi_2I_9$ absorbers were shown to be the most effective and promising. These theoretical outcomes agree with the previously published study.[84]

3.2.4. Effect of Series Resistance. Figure 11 shows the effect of series resistance (R_S) on the performance of the $Cs_3Bi_2I_9$-based PSC, with R_S varying from 0 to 6 Ω-cm^2 and shunt resistance (R_{SH}) fixed at 10^5 Ω-cm^2, respectively. Compared to V_{OC} and J_{SC}, the *FF* and PCE were determined to be the metrics that were more affected by varying R_S. Figures 11(a) and (b) show that while the V_{OC} and J_{SC} were not affected significantly by the increase in R_S from 0 to 6 Ω-cm^2 across all of the studied devices, on the contrary, the *FF* reduced dramatically. Maximum V_{OC} and J_{SC} were observed for the ITO/SnO_2/$Cs_3Bi_2I_9$/Cu_2O/Au and ITO/TiO_2/$Cs_3Bi_2I_9$/Cu_2O/Au PSCs, while the ITO/PCBM/$Cs_3Bi_2I_9$/Cu_2O/Au PSC configuration showed poor results. This sort of outcome was very much similar to that in the

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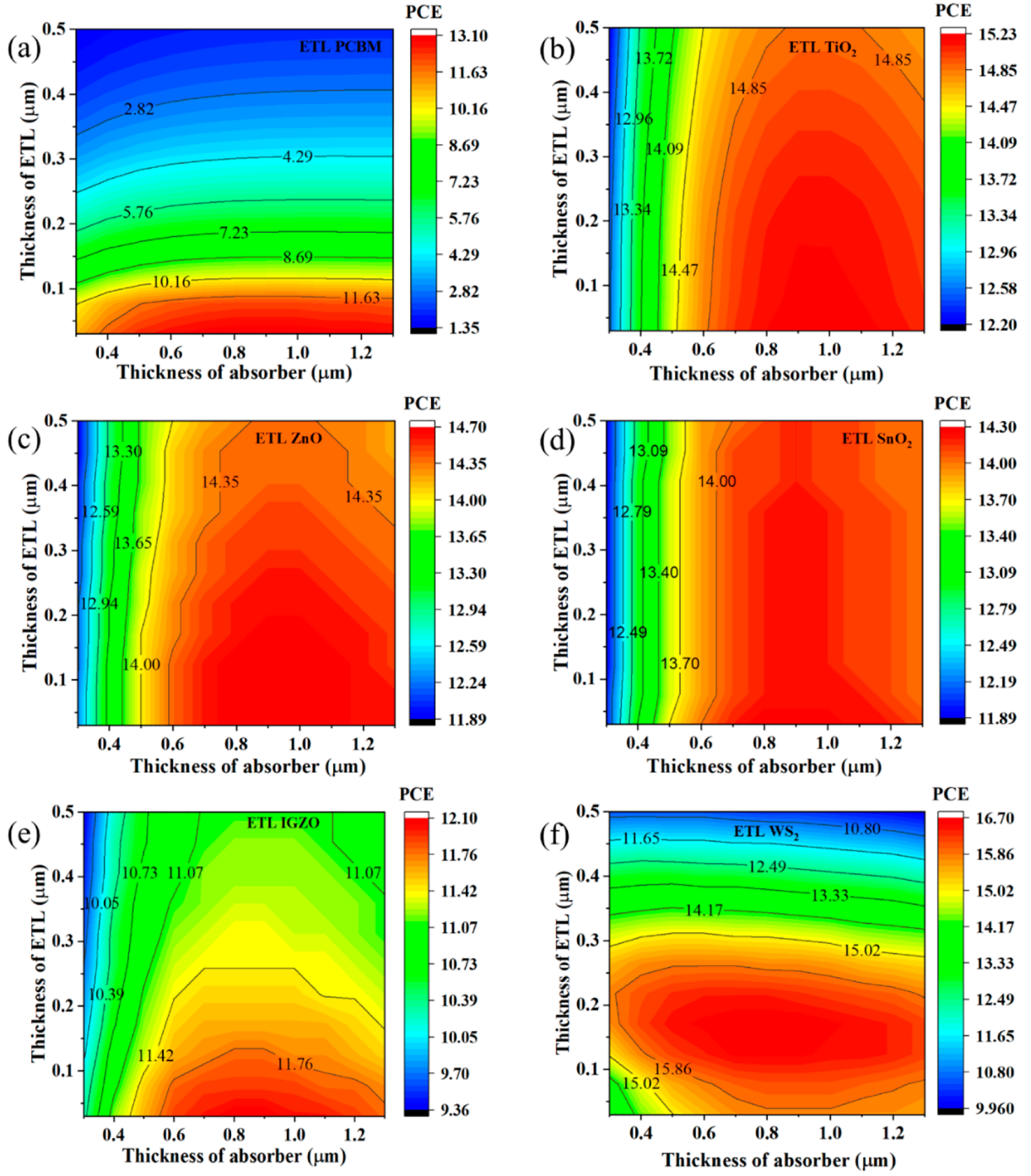


Figure 10. Contour plots of PCE with a simultaneous variation of the thickness of the $Cs_3Bi_2I_9$ absorber and ETLs like (a) PCBM, (b) TiO_2, (c) ZnO, (d) SnO_2, (e) IGZO, and (f) WS_2.

previous literature where V_{OC} and J_{SC} were not affected much.[85] The PCEs of these six studied structures were greatly influenced by an increase in R_S, as the *FF* was inversely proportional to the R_S. The higher R_S results in more resistive losses and reduces the squareness of the *J*−*V* curve, i.e., reduces the *FF*. The impact of R_S on solar cell characteristics is explained with the help of eqs 8 and 9.

$$I_{SC} = I_0(e^{qV_{OC}/nKT} - 1) \tag{8}$$

$$I_{SC} = I_L - I_0(e^{qV_{OC}/nKT} - 1) - \frac{V_{OC} + I_{SC}R_S}{R_{SH}} \tag{9}$$

Here I_L and I_{SC}, respectively, signify the light induced current and short circuit current. The formula above makes it clear that when R_S rises, the I_{SC} value decreases. It is mostly responsible for the decline in PCE and *FF*.[86,87]

3.2.5. Effect of Shunt Resistance. Figure 12 specifies the correlation between the values of the PV parameters V_{OC}, J_{SC}, *FF*, and PCE and the range of R_{SH}, which is extended from 10^1 to 10^7 Ω-cm^2 across all six desirable SC configurations for a fixed R_S. With rising R_{SH}, the V_{OC}, *FF*, and PCE values all followed the

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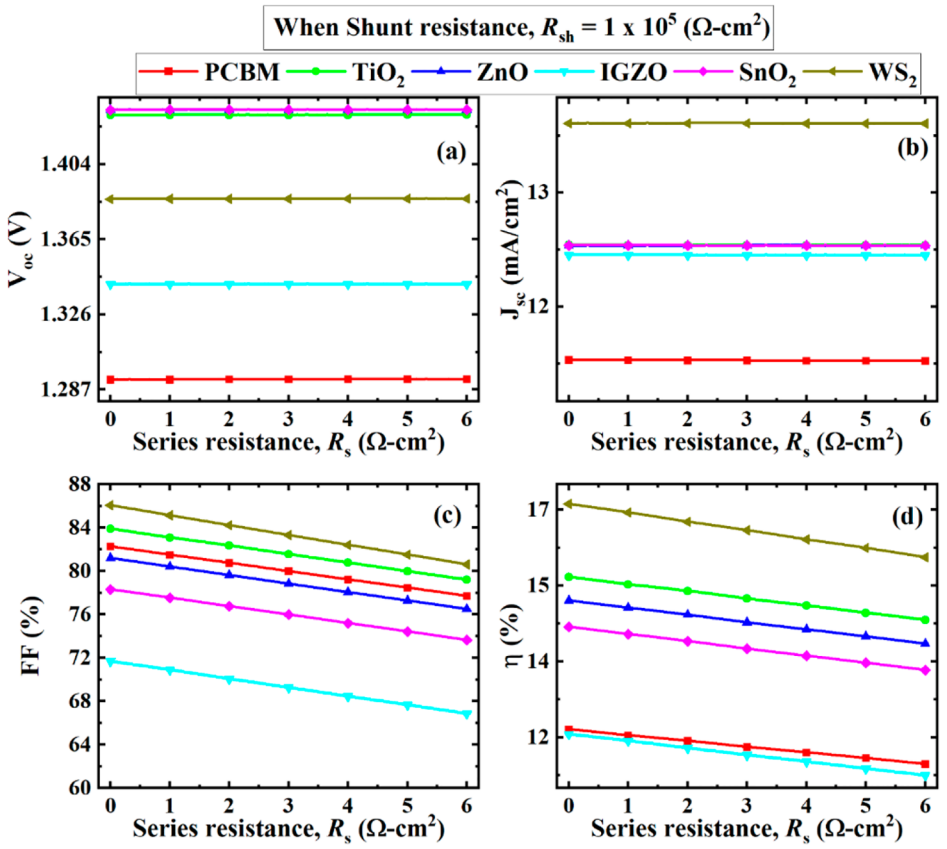


Figure 11. Effect of R_S on (a) V_{OC}, (b) J_{SC}, (c) *FF*, and (d) PCE for ITO/ETL/$Cs_3Bi_2I_9$/Cu_2O/Au configurations.

same trend except for J_{SC}. And from 10^1 Ω-cm^2 to 10^3 Ω-cm^2, PV performance indicators V_{OC}, *FF*, and PCE surged up instantly and subsequently saturated with further increases in R_{SH}.

In the case of V_{OC}, the ITO/SnO_2/$Cs_3Bi_2I_9$/Cu_2O/Au PSC showed a maximum of ∼1.32 V constantly after 10^3 Ω-cm^2, while the ITO/PCBM/$Cs_3Bi_2I_9$/Cu_2O/Au PSC exhibited a minimum value of ∼1.23 V in comparison to the other PSCs (Figure 12(a)). The J_{SC} of all of the specifications is almost unchanged where the ITO/WS_2/$Cs_3Bi_2I_9$/Cu_2O/Au and ITO/PCBM/$Cs_3Bi_2I_9$/Cu_2O/Au PSCs showed a maximum and minimum of ∼13.75 and ∼11.4 mA/cm^2, respectively (Figure 12(b)). Among all of the specifications, the *FF* of the ITO/WS_2/$Cs_3Bi_2I_9$/Cu_2O/Au PSC was the highest at ∼85% and the ITO/IGZO/$Cs_3Bi_2I_9$/Cu_2O/Au PSC showed the lowest value of ∼68% (Figure 12(c)). A similar trend of PCE was visible while the ITO/WS_2/$Cs_3Bi_2I_9$/Cu_2O/Au PSC relatively outperformed with ∼16.7% efficiency, whereas the ITO/IGZO/$Cs_3Bi_2I_9$/Cu_2O/Au PSC showed the lowest PCE of ∼12% (Figure 12(d)). R_{SH} is typically caused by imperfections introduced into the PSCs during the manufacturing process. At a large enough R_{SH}, the p-n junction can facilitate junction current flow with little resistance.[46] The trend of variation observed with the different PV parameters as a result of the change in R_{SH} was in line with the previous study.[85]

3.2.6. Effect of Temperature. Knowledge of PV performance at high working temperatures is crucial for predicting the lifespan of SCs without significant degradation in PCE. When overheated, the layers of most SC layouts become defect-prone, causing performance fluctuations. Figure 13 depicts the consequence of temperature fluctuations on the performance criteria for the six specifications, including V_{OC}, J_{SC}, *FF*, and PCE, for $Cs_3Bi_2I_9$-based PSCs. It is also worth noting that the variation in temperature as a function of time is not currently supported in the simulator under consideration. Therefore, the effect of different temperatures on PSCs was listed in the paper as a general overview, without considering the time dimensions, and is the main limitation of the temperature model considered in this work.

The V_{OC} values of all of the studied $Cs_3Bi_2I_9$-based PSCs decline with the increase of temperature as well as produce enhanced reverse saturation current. The V_{OC} of the ITO/PCBM/$Cs_3Bi_2I_9$/Cu_2O/Au PSC decreased from 1.355 to 0.827 V, which was the lowest among all of the PSCs. On the contrary, the ITO/TiO_2/$Cs_3Bi_2I_9$/Cu_2O/Au PSC was superior among all of the specifications, which showed a decline of V_{OC} from 1.487 to 0.998 V (Figure 13(a)). With the enhancement of temperature, the J_{SC} of the specifications was almost unchanged except for the ITO/PCBM/$Cs_3Bi_2I_9$/Cu_2O/Au PSC. It was evident that from 275 to 475 K, the bandgap of the absorber is not substantially affected by the rising temperature, which reflects a similarity with the performance in the previous literature.[88,89] The *FF* of the studied specifications decreased with the increase of temperature except for the ITO/IGZO/$Cs_3Bi_2I_9$/Cu_2O/Au PSC, which tends to decrease from 300 K and at 350 K again tends to increase as well. When it comes to PCE, however, the increase in temperature tends to have an adverse effect with declining PCE for all of the devices.

3.2.7. Effect of Capacitance and Mott−Schottky. Figure 14(a) demonstrates the influence of capacitance in the voltage range of −0.5 to 0.8 V and a stable frequency range of 1 MHz for six distinct combinations of $Cs_3Bi_2I_9$-based PSCs with various ETLs. As seen in Figure 14(a), as the supplied voltage is

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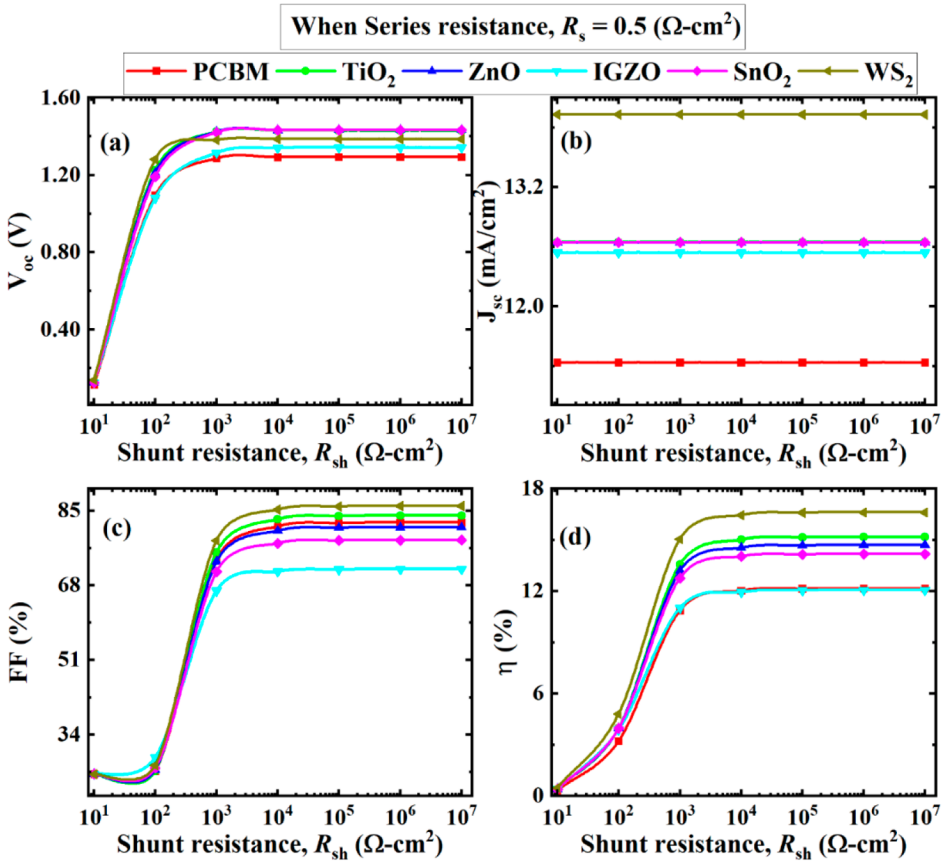


Figure 12. Effect of R_{SH} on (a) V_{OC}, (b) J_{SC}, (c) *FF*, and (d) PCE for ITO/ETL/$Cs_3Bi_2I_9$/Cu_2O/Au configurations.

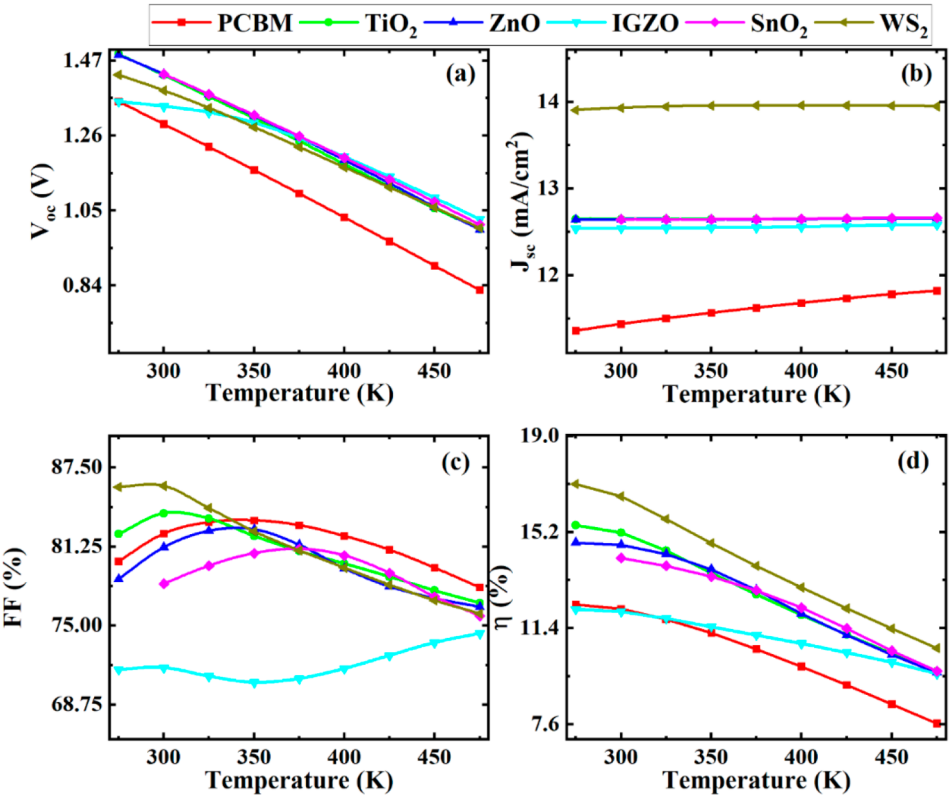


Figure 13. Effect of operating temperature on (a) V_{OC}, (b) J_{SC}, (c) *FF*, and (d) PCE for ITO/ETL/$Cs_3Bi_2I_9$/Cu_2O/Au configurations.

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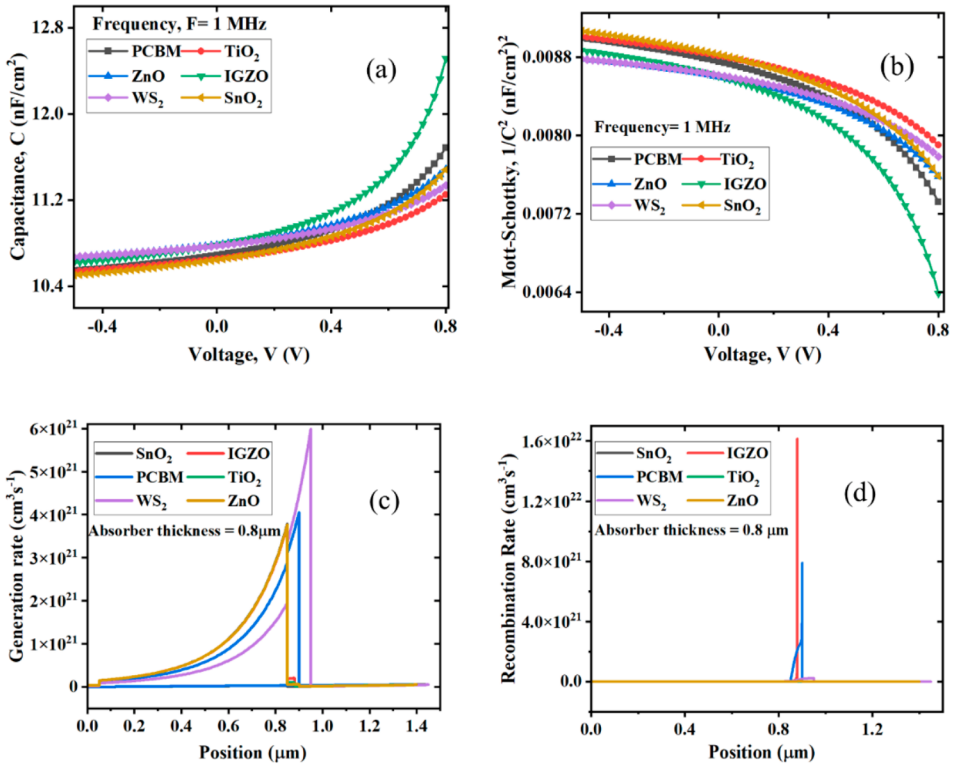


Figure 14. (a) Capacitance (C), (b) Mott−Schottky ($1/C^2$), (c) generation rate, and (d) recombination rate of $Cs_3Bi_2I_9$-based PSCs with six different ETLs.

increased, the capacitance rises exponentially until the saturation point is reached. With the increase of voltage from −0.5 V, the capacitance of the specifications tends to increase. While the ITO/IGZO/$Cs_3Bi_2I_9$/Cu_2O/Au PSC showed a capacitance of 12.5 C, the capacitance for other specifications ranges from 11 to 11.5 C. The PSCs enter a depletion situation in the absence of a bias because, when an upward bias of about 0.5 V is applied, the width of the depletion region shrinks to become almost equal to the thickness of the absorber layer. For this reason, increasing the forward bias voltage results in an increase in capacitance while maintaining the Mott−Schottky (MS) relationship. In contrast to previous observations, only during voltage fluctuations, the current exceeds the saturation current of the contacts. As voltage is reduced, the current is also significantly diminished.

For six different configurations of $Cs_3Bi_2I_9$-based PSCs with listed ETLs, Figure 14(b) displays the impact of MS features over an operating voltage of −0.5 to 0.8 V alongside a consistent frequency band of 1 MHz. Here the ITO/TiO_2/$Cs_3Bi_2I_9$/Cu_2O/Au PSC exhibited the highest MS value of 0.080 ($1/C^2$), while the ITO/IGZO/$Cs_3Bi_2I_9$/Cu_2O/Au PSC showed the lowest MS value of 0.0064 ($1/C^2$). The widely applied MS method can be used to estimate the built-in potential (V_{bi}). Although the outcomes are worse owing to the various work activities of the metal electrodes, the central emphasis of the MS principle is on the characteristics of the p-n junction, in which the intercept point on the x-axis of the $C-V$ curve is equal to the V_{bi} of the respective junction. The MS attributes for a given device decrease whenever the V_{bi} values rise. Here, the simulation results for every device seemed to be consistent across all key metrics, making them easily equivalent and consistent with the observations in previous reports.[90,91]

3.2.8. Effect of Generation and Recombination Rate. Figures 14(c) and (d) project the carrier generation as well as recombination rates for six $Cs_3Bi_2I_9$-based PSCs throughout a range of 0−1.5 μm, with all the other input variables remaining unchanged. An electron is uplifted from the valence to the conduction band due to the absorption of a photon of energy equal to or greater than the bandgap of the absorber material, creating a hole in the valence band. Such a process generates electron−hole pairs in all p-n junction devices. Between 0.8 and 1.0 μm, the generation rates of all of the configurations attain peak values, with the ITO/WS_2/$Cs_3Bi_2I_9$/Cu_2O/Au device exhibiting the highest rate at 6×10^{21} cm^3 s^{-1} of all of the devices studied. SCAPS-1D calculates the number of electron−hole pairs produced, $G(x)$, based on the photon flux $N_{phot}(\lambda,x)$, which is then used in eq 10 to compute the value of $G(x)$ at each wavelength range and location.

$$G(\lambda, x) = \alpha(\lambda, x) \cdot N_{phot}(\lambda, x) \tag{10}$$

In parallel, holes and electrons inside the conduction band recombine with a particular recombination rate, which is the inverse of the generation rate. The recombination rate in PSCs is influenced by the charge carrier density and lifetime. The decrease in electron−hole recombination is due to the presence of defect states in the buffer layer. ITO/IGZO/$Cs_3Bi_2I_9$/Cu_2O/Au ETL had the highest recombination peak among the specifications evaluated. However, all of the devices showed recombination peaks between 0.8 and 0.9 μm. Electron−hole recombination manifests in the devices due to the creation of an energy state in the valence−conduction band's ideal range. Grain boundaries and defects at junctions and interfaces can lead to an unequal distribution of the recombination rate in PSCs.[90]

3.2.9. Effect of Current−Voltage Density and Quantum Efficiency. Figure 15(a) depicts the current−voltage density ($J-$

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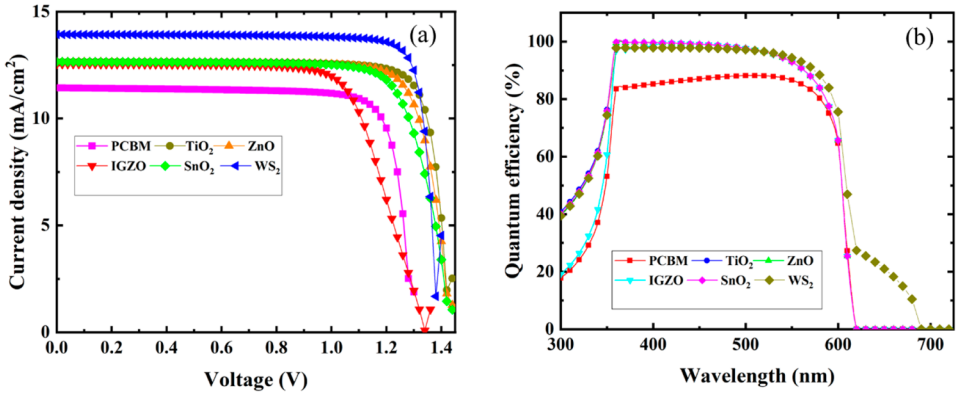


Figure 15. (a) *J*–*V* and (b) *QE* features for six different configurations of $Cs_3Bi_2I_9$-based PSCs.

Table 5. Comparison of PV Parameters of $Cs_3Bi_2I_9$-Based PSCs

Type[a]	Cell structure	V_{OC} (V)	J_{SC} (mA/cm^2)	*FF* (%)	PCE (%)	Ref
1	Glass/FTO/ETL/$Cs_3Bi_2I_9$/HTL/Au	0.92	22.07	68.21	13.82	8
1	Glass/FTO/TiO_2/$Cs_3Bi_2I_9$/PTAA/Au	-	-	-	8.00	94
1	FTO/TiO_2/mp-TiO_2/$Cs_3Bi_2I_9$/P3HT/Ag	0.02	0.18	0.37	0.02	95
1	FTO/TiO_2/mp-TiO_2/$Cs_3Bi_2I_9$/Spiro-OMeTAD/Ag	0.85	2.15	0.60	1.09	96
2	ITO/PCBM/$Cs_3Bi_2I_9$/Cu_2O/Au	1.292	11.44	82.27	12.16	b
2	ITO/TiO_2/$Cs_3Bi_2I_9$/Cu_2O/Au	1.429	12.65	83.91	15.17	b
2	ITO/ZnO/$Cs_3Bi_2I_9$/Cu_2O/Au	1.432	12.64	81.18	14.70	b
2	ITO/C_{60}/$Cs_3Bi_2I_9$/Cu_2O/Au	1.213	9.77	82.56	9.78	b
2	ITO/IGZO/$Cs_3Bi_2I_9$/Cu_2O/Au	1.342	12.54	71.68	12.06	b
2	ITO/SnO_2/$Cs_3Bi_2I_9$/Cu_2O/Au	1.432	12.64	78.3	14.18	b
2	ITO/WS_2/$Cs_3Bi_2I_9$/Cu_2O/Au	1.386	13.93	86.03	16.61	b

[a]Note: 1, experimental; 2, theoretical. [b]This work.

V) characteristics for six distinct ETLs for an ITO/ETL/$Cs_3Bi_2I_9$/Cu_2O/Au device configuration. In contrast to all of the PSCs, the ITO/PCBM/$Cs_3Bi_2I_9$/Cu_2O/Au structure generated the least photocurrent, whereas the ITO/WS_2/$Cs_3Bi_2I_9$/Cu_2O/Au structures exhibited the highest photocurrent. From the figure, it is evident that a bandgap above 1.8 eV is not preferable for the desired PV performance with the given configuration. The ETL band configuration influences the flow of photogenerated current in $Cs_3Bi_2I_9$-based PSCs, confirming prior findings.[91,92]

The quantum efficiency (*QE*) plots for all of the studied devices are shown in Figure 15(b). It can be observed that the *QE* decreases after a wavelength of 600 nm, which is the wavelength corresponding to the bandgap of ∼2.03 eV for the $Cs_3Bi_2I_9$ absorber. The *QE* plots peaked at ∼350 nm for all of the ETLs. *QE* curves for the ETLs except PCBM are almost of a similar nature. The lower *QE* for PCBM may be due to the lesser photon absorption and interface characteristics exhibited by the PCBM polymeric layer with a $Cs_3Bi_2I_9$ absorber.[93] The WS_2 ETL with a narrower band gap of 1.8 eV allows for greater photon uptake in the $Cs_3Bi_2I_9$ absorber, leading to a larger photocurrent density.

3.3. Comparison of SCAPS-1D Results with Previous Work. In comparison to recent theoretical and experimental studies on lead-free $Cs_3Bi_2I_9$-based PSCs with a different configuration, Table 5 compares our findings. To date, PSCs with the Glass/FTO/ETL/$Cs_3Bi_2I_9$/HTL/Au specification exhibited the highest experimental efficiency of 13.82%, with a V_{OC} of 0.92 V, J_{SC} of 22.07 mA/cm^2, and *FF* of 68.21%. In the theoretical investigation being conducted, ITO/WS_2/$Cs_3Bi_2I_9$/Cu_2O/Au exhibited a maximal PCE of about 16.61% with V_{OC} of 1.39 V, J_{SC} of 13.93 mA/cm^2, and *FF* of 86.03%. Other theoretical configurations of this study like ITO/TiO_2/$Cs_3Bi_2I_9$/Cu_2O/Au, ITO/ZnO/$Cs_3Bi_2I_9$/Cu_2O/Au, and ITO/SnO_2/$Cs_3Bi_2I_9$/Cu_2O/Au also showed significant PCEs of about 15.17%, 14.70%, 14.18%, respectively, which was a notable factor in comparison to all of the past experimental literature.

4. SUMMARY

The study presented here aimed to investigate the feasibility of developing Pb-free PV cells utilizing a $Cs_3Bi_2I_9$ absorber. Through extensive numerical simulations using the SCAPS-1D tool and detailed computation of the optoelectronic and structural properties of the absorber using DFT, we have explored 49 possible configurations with different ETLs and HTLs. Our investigation has led to several key findings provided below:

(1) The confirmation of a 2.453 eV indirect bandgap for the hexagonal structure of $Cs_3Bi_2I_9$ is consistent with previous reports and demonstrates its potential for solar cell device applications. Our analysis of the charge distribution revealed isotropic behavior along crystallographic planes, and we identified ionic and covalent bonding between Cs, I and Bi, I atoms, respectively. Optical data indicated that $Cs_3Bi_2I_9$ has strong absorption coefficients in both the IR and visible regions, making it a promising contender for optoelectronic applications.

(2) Analysis of different ETL and HTL combinations led to the identification of ITO/WS_2/$Cs_3Bi_2I_9$/Cu_2O/Au and ITO/C_{60}/$Cs_3Bi_2I_9$/Cu_2O/Au as having the highest and lowest PCEs (16.61% and 9.78%, respectively) among the studied PSC

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structures. Other ETLs such as SnO_2, ZnO, and TiO_2 exhibited PCEs in the range of 14−15%.

(3) The optimization procedure revealed that the ideal thicknesses for the absorber and ETL were 800 and 150 nm, respectively. Additionally, we found that Cu_2O was the best-performing HTL due to its direct bandgap nature, nontoxicity, and compatibility with different organic/inorganic layers in SCs.

(4) We investigated the impact of series and shunt resistances and operating temperatures on PV parameters, with consistent results in line with basic device equations. We found that the performance of the PSC was impacted by both series and shunt resistance parameters and that PSCs performed optimally at temperatures below 313 K, with efficiency rapidly declining at higher temperatures.

(5) We thoroughly investigated the capacitance, Mott−Schottky, generation, and recombination rates, *J*−*V*, and *QE* characteristics of the optimized structures, offering a deeper understanding of the underlying phenomena guiding the operation of these devices.

This systematic investigation provides researchers working in the field of Bi halide-based PSCs with a valuable resource for designing cost-effective PSCs with suitable ETL and HTL combinations, potentially saving significant time and resources.

■ ASSOCIATED CONTENT

Data Availability Statement

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

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Notes

The authors declare no competing financial interest.

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